

UNI/ELECENG 3TP3/ELECENG 3TP3.MD

UNI/ELECENG 3TP3/EXAMPLES/IMPULSE-RESPONSE-LONG.MD

UNI/ELECENG 3TP3/EXAMPLES/GRAPHICAL-CONVOLUTION.MD

$x(t)$ is the size of a signal.

Energy E_x of a signal $x(t) = \int_{-\infty}^{\infty} x(t)^2 dt$ is $E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$

If the energy is infinite, signal power $P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)^2 dt$. If signal is complex, $P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} |x(t)|^2 dt$. For power to exist, the signal needs regularity.

Analog signals: All values are possible, continuous-time.

Digital signals: Only certain values are possible, continuous-time.

Analog and discrete-time: Any value, $x(t)$ exists only at certain times

Digital and discrete-time: Only certain values, $x(t)$ exists only at certain times

Periodic signals: $x(t) = x(t + T_0)$ for all t . The period is the smallest value of T_0 that satisfies the equation

Casual signals: $x(t) = 0$ for $t < 0$

Non-casual signals: $x(t)$ that starts before $t = 0$

Anti-casual signals: $x(t) = 0$ for $x \geq 0$

Energy signals: Signals with finite energy

Power signals: Signals with finite and non-zero power

..

An energy signal cannot be a power signal and vice versa, but a signal can also be neither.

Shifting: $\phi(t) = x(t - T)$: shifted version of $x(t)$, $T > 0$ is right shift (time delay), $T < 0$ is left shift (time advance).

Shift and time scaling: $\phi(t) = x(at - b)$ Two options: time shift by b , then time-scale by a . Or time-scale by a , then shift by $\frac{b}{a}$.

Odd and even components of a signal: $x_e(t) = \frac{1}{2}[x(t) + x(-t)] + \frac{1}{2}[x(t) - x(-t)]$, first brackets are even, second is odd

Time-invariant system: system parameters do not change with time. Shift in T in input = shift by T in output

Casual systems: Output at t_0 depends only on put up to time t_0

Memoryless systems: output at time t_0 depends only on input at time t_0

Finding response $y(t)$ of a system to input $x(t)$: response = output due to initial conditions + output due to input, and total response = zero-input response + zero-state response

Zero-input response is the output when input $x(t) = 0$, zero-state response is the output when initial conditions are zero. The solutions are the roots of the polynomial $Q(\lambda) = 0$. There are N distinct solutions for $y_0(t)$ when $x(t) = 0$, $c_N e^{\lambda_N t}$. The formula $Q(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots = 0$ is the characteristic equation. $\lambda_1, \lambda_2, \dots$ are the eigenvalues or characteristic roots of the system. $e^{\lambda_i t}$ are the modes of the system. Zero-input response is a linear combination of the modes.

By splitting the input $x(t)$ to a series of small time intervals and rectangular pulses, through superposition, the output $y(t)$ is a sum of the individual responses. As the time interval becomes small, the rectangles become impulses. Then the input can be viewed as a continuous sum of time-shifted impulses

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

The impulse response is the response $y(t) = h(t)$ when the input $x(t)$ is made the dirac impulse $\delta(t)$. The impulse happens at $t = 0$ but the system can keep responding past $t = 0$. In general, the form of

$$h(t) = b_0 \delta(t) + (\text{characteric modes})u(t)$$

Finding impulse response:

- Find characteristic modes (response for $t > 0$ and thus input is zero). This means the RHS is zero.
- Add b_0
- Try to match coefficients of $\delta(t)$ to $h(t)$ from the LHS to RHS. If there are derivatives of the input $\delta(t)$, use the property $h'(t) \supset K_1 \delta(t)$.
 - This property comes from the fact that $h(t)$ has a jump at $t = 0$ of size $h(0^+) - h(0^-) = K_1 - 0 = K_1$. A jump is the size of the jump multiplied by a Dirac delta, hence the derivative will contain the jump term. Differentiate again and get $h'' \supset K_1 \delta'(t)$. There is also a possible jump in $h'(t)$. If $K_2 = h'(0^+)$, following the above, $h'' \supset K_2 \delta(t)$. Since h'' now contains two terms, $h'' \supset K_1 \delta'(t) + K_2 \delta(t)$.
 - Notice how the $h(t)$ term does not have any $\delta(t)$ terms inside it.
- Matching coefficients gives the values for $h^N(0^+)$. Since the characteristic modes are the response for $t > 0$, substitute $t = 0^+$ to find an expression for $h(0^+)$. Differentiate the characteristic modes to get more equations for $h^N(0^+)$ as functions of c_n .
- Solve for the constants and you now have $h(t)$. Don't forget the $u(t)$ on the characteristic modes.
- **IMPULSE-RESPONSE-LONG**

Alternatively, identifying $Q(D)y(t) = P(D)x(t)$:

- $h(t) = b_0\delta(t) + [P(D)y_n(t)]u(t)$ where $y_n(t)$ is the linear combination of the characteristic modes.
- That's it

Going back to approximating the input with rectangles, each rectangle (called a pulse $p(t)$) has amplitude of $x(t)$ and a narrow width of $\Delta\tau$. The signal can thus be reconstructed by shifting the n -th pulse to the n -th interval and scaling by $x(t)$:

$$x(t) \approx \sum_n x(n\Delta\tau)p(t - n\Delta\tau)$$

Multiply this by $\frac{\Delta\tau}{\Delta\tau}$.

$$x(t) = \lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau) \left[\frac{p(t - n\Delta\tau)}{\Delta\tau} \right] \Delta\tau$$

The term $\frac{p(t-n\Delta\tau)}{\Delta\tau}$ approximates to $\delta(t)$ as $\Delta\tau \rightarrow 0$. Since by definition $\delta(t) \rightarrow h(t)$, and the properties of an LTI system mean all the scaling, shifting, and superposition applied to $\delta(t)$ will apply the same way to $h(t)$:

$$\lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau)\delta(t - n\Delta\tau)\Delta\tau \longrightarrow \lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau)h(t - n\Delta\tau)\Delta\tau$$

As $\Delta\tau \rightarrow 0$, $n\Delta\tau$ turns into the continuous τ , $\Delta\tau$ becomes $d\tau$, and the summation becomes a continuous integral sum.

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau$$

This is otherwise known as convolution, $y(t) = x(t) * h(t)$.

If you know a LTI system's impulse response $h(t)$, you can find its zero state output $y(t)$ for any $x(t)$ through convolution.

Convolution is:

- Commutative: $x_1(t) * x_2(t) = x_2(t) * x_1(t)$
- Distributive: $x_1(t) * [x_2(t) + x_3(t)] = x_1(t) * x_2(t) + x_1(t) * x_3(t)$
- Associative: $x_1(t) * [x_2(t) * x_3(t)] = [x_1(t) * x_2(t)] * x_3(t)$
- Shiftable: if $x_1(t) * x_2(t) = y(t)$:
 - $x_1(t) * x_2(t - T) = x_1(t - T) * x_2(t) = y(t - T)$
 - or $x_1(t - T_1) * x_2(t - T_2) = y(t - T_1 - T_2)$

Convolution with an impulse: $x(t) * \delta(t) = \int x(\tau)\delta(t - \tau) d\tau$

Width property: The width of $x_1(t) * x_2(t)$ is the sum of the two finite widths $T_1 + T_2$ of the two convolved functions.

For two casual signals:

$$\begin{aligned} y(t) &= x(t) * h(t) = \left[\int_0^t x(\tau) h(t - \tau) d\tau \right] u(t) \\ &= \left[\int_0^t h(\tau) x(t - \tau) d\tau \right] u(t) \end{aligned}$$

Graphical method of convolution:

- This method is easiest when the signals are easy to sketch.
- Recall $c(t) = \int_{-\infty}^{\infty} x(\tau) g(t - \tau) d\tau$.
- Given the convolution we are trying to solve has the form $x(t) * g(t) = c(t)$:
 - Substitute $t = \tau$ into $x(t)$ for $x(\tau)$
 - Substitute $t = \tau$ then flip $g(t)$ across y-axis for $g(-\tau)$. Mark $g(-\tau)$'s left and right edges.
 - Shift by an amount t . The edges of $g(-\tau)$ will become $\text{offset}_1 + t < g(\tau) < \text{offset}_2 + t$.
 - A 'region' begins whenever an edge of the moving g signal hits an edge of the stationary x signal and ends right before another edge hit occurs. Break the entire convolution up into regions of t .
 - Inside each region, the limits of integration become $\tau_{\text{lower}} = \max(\text{left edge of } x, \text{left edge of moving } g)$, $\tau_{\text{upper}} = \min(\text{right edge of } x, \text{right edge of moving } g)$.
 - Evaluate $\int_{\tau_{\text{lower}}}^{\tau_{\text{upper}}} x(\tau) g(t - \tau) d\tau$ symbolically to get the formula for that interval.
 - If the signals are simple enough, eg. pulses or ramps, you can just vibrate it out by solving for $c(t)$ at boundary t values and connecting the points with straight lines.
- GRAPHICAL-CONVOLUTION

Output of a system for exponential input:

$$\begin{aligned} y(t) &= h(t) * e^{st} \\ &= \int h(\tau) e^{s(t-\tau)} d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau \\ &= H(s) e^{st} \quad \left(H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau \right) \\ \therefore H(s) &= \frac{y(t)}{e^{st}} = \frac{\text{output}}{\text{input}} \end{aligned}$$

And so $H(s)$ is the **transfer function**.

BIBO stability: bounded input and bounded output. $\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$. BIBO stable systems' impulse responses are absolutely integrable.

Asymptotic stability: as $t \rightarrow \infty$, all natural modes $e^{\lambda_i t} \rightarrow 0$.

- Stable: If all $\text{Re}(\lambda_i) < 0$, the mode will decay to zero as $t \rightarrow \infty$.
- Unstable: If at least one $\text{Re}(\lambda_i) > 0$, the mode will grow as $t \rightarrow \infty$ and such the mode is unstable.
 - The system will also be unstable if the roots are repeated and imaginary, since repeated imaginary roots can cause terms like $te^{\lambda t}$ which grow instead of oscillating.
- Asymptotically unstable: Both roots are imaginary and different (will oscillate).

Impulse Response (Method 1)

Find the impulse response of the zero-initial-condition system

$$y'''(t) + 6y''(t) + 11y'(t) + 6y(t) = x'(t) + 2x(t).$$

Here the input-side constant coefficient is $b_0 = 2$. For an impulse input, $x(t) = \delta(t)$, the output is $h(t)$ and

$$h'''(t) + 6h''(t) + 11h'(t) + 6h(t) = \delta'(t) + 2\delta(t).$$

Solution

1. Find the characteristic modes

For $t > 0$, the impulse forcing is zero, so we want to solve the homogeneous equation derived from the LHS:

$$\begin{aligned} r^3 + 6r^2 + 11r + 6 &= 0 \\ (r+1)(r+2)(r+3) &= 0 \end{aligned}$$

Therefore, for $t > 0$,

$$h(t) = c_1 e^{-t} + c_2 e^{-2t} + c_3 e^{-3t}$$

For the causal impulse response, write this as

$$h(t) = f(t)u(t), \quad f(t) = c_1 e^{-t} + c_2 e^{-2t} + c_3 e^{-3t}$$

2. Match the impulse terms

For $h(t) = f(t)u(t)$, recall the terms of its derivatives:

$$\begin{aligned} h' &\supset f(0^+)\delta, \\ h'' &\supset f(0^+)\delta' + f'(0^+)\delta, \\ h''' &\supset f(0^+)\delta'' + f'(0^+)\delta' + f''(0^+)\delta \end{aligned}$$

Therefore, the singular terms on the left-hand side are

$$\begin{aligned} \delta'' &: f(0^+), \\ \delta' &: f'(0^+) + 6f(0^+), \\ \delta &: f''(0^+) + 6f'(0^+) + 11f(0^+) \end{aligned}$$

The right-hand side is $\delta' + 2\delta$, so matching coefficients gives

$$\begin{cases} f(0^+) = 0, \\ f'(0^+) + 6f(0^+) = 1, \\ f''(0^+) + 6f'(0^+) + 11f(0^+) = 2 \end{cases}$$

Hence:

$$f(0^+) = 0, \quad f'(0^+) = 1, \quad f''(0^+) = -4$$

3. Solve

Using the characteristic-mode expression, its derivative, and second derivative then substituting 0^+ ,

$$\begin{cases} c_1 + c_2 + c_3 = 0 \\ -c_1 - 2c_2 - 3c_3 = 1 \\ c_1 + 4c_2 + 9c_3 = -4 \end{cases}$$

which gives

$$c_1 = \frac{1}{2}, \quad c_2 = 0, \quad c_3 = -\frac{1}{2}$$

End

Thus the impulse response is

$$h(t) = \frac{1}{2}(e^{-t} - e^{-3t})u(t)$$

Graphical Convolution

Find $c(t) = x(t) * g(t)$

Given:

- $x(t) = t$ for $0 \leq t \leq 2$
- $g(t) = 1$ for $0 \leq t \leq 1$

Solution

Step 1: Set up the boundaries

The signal $x(\tau) = \tau$ is defined on the fixed interval $[0, 2]$.

The original pulse $g(\tau)$ is nonzero for $0 \leq \tau \leq 1$. After flipping, $g(-\tau)$ is nonzero for $-1 \leq \tau \leq 0$. After shifting, $g(t - \tau)$ is nonzero for $t - 1 \leq \tau \leq t$.

- Trailing edge: $\tau = t - 1$
- Leading edge: $\tau = t$
- Inside the interval: $g(t - \tau) = 1$

Step 2: Determine the critical transition points for t

Set the moving edges equal to the stationary boundaries, $\tau = 0$ and $\tau = 2$:

- Leading edge enters the stationary window: $t = 0$
- Trailing edge enters the stationary window: $t - 1 = 0 \implies t = 1$
- Leading edge leaves the stationary window: $t = 2$
- Trailing edge leaves the stationary window: $t - 1 = 2 \implies t = 3$

The timeline divides into five regions:

- $t < 0$
- $0 \leq t < 1$
- $1 \leq t < 2$
- $2 \leq t < 3$
- $t \geq 3$

Step 3: Compute the integral for each region

Region 1: $t < 0$: No overlap

$$c(t) = 0$$

Region 2: $0 \leq t < 1$: Pulse enters the ramp

The overlap is $\tau \in [0, t]$, and the integrand is

$$x(\tau)g(t - \tau) = \tau \cdot 1 = \tau$$

Therefore,

$$c(t) = \int_0^t \tau d\tau = \left[\frac{1}{2} \tau^2 \right]_0^t = \frac{1}{2} t^2$$

Region 3: $1 \leq t < 2$: Pulse is completely inside the ramp

The overlap is $\tau \in [t-1, t]$. The integrand is τ , so

$$c(t) = \int_{t-1}^t \tau d\tau = \left[\frac{1}{2} \tau^2 \right]_{t-1}^t = \frac{1}{2} (t^2 - (t-1)^2) = t - \frac{1}{2}$$

Notice how the result is linear because a window of fixed width is sliding under a straight ramp.

Region 4: $2 \leq t < 3$: Pulse exits the ramp

The overlap is $\tau \in [t-1, 2]$. Thus,

$$c(t) = \int_{t-1}^2 \tau d\tau = \left[\frac{1}{2} \tau^2 \right]_{t-1}^2 = \frac{1}{2} (4 - (t-1)^2) = -\frac{1}{2} t^2 + t + \frac{3}{2}$$

Region 5: $t \geq 3$: Complete exit

Finally,

$$c(t) = 0$$

End

Therefore the result, when pieced back together, is

$$c(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{2} t^2, & 0 \leq t < 1, \\ t - \frac{1}{2}, & 1 \leq t < 2, \\ -\frac{1}{2} t^2 + t + \frac{3}{2}, & 2 \leq t < 3, \\ 0, & t \geq 3 \end{cases}$$

You can do a check and verify that this solution is perfectly continuous on t .